# ■■ Road Surface Anomaly Detection

Comprehensive Sensor Analysis Report

Session Date: 26 February 2026 | Time Range: 13:14 – 13:39

System: Live Ultrasonic Sensor @ 10.83.97.128:8501

## Executive Summary

This report presents a detailed analysis of **50 road surface anomaly events** captured by an ultrasonic distance sensor mounted on a vehicle during a monitoring session on 26 February 2026 (13:14 – 13:39 IST). The system detected a mix of potholes, speed bumps, and surface irregularities using real-time deviation analysis against a rolling baseline distance.

Key findings: **19 Deep Potholes**, **3 Shallow Potholes**, and **28 Speed Bumps** were identified. **31 events (62%)** were classified as **Deep/Dangerous** — requiring urgent road repair. The average pothole depth was **23.4 cm**, with a maximum recorded depth of **69.7 cm**. Thread errors (237 of 3030 frames) suggest occasional sensor noise that should be investigated.

### Key Statistics

| Metric | Value |
|---|---|
| Total Detections | 50 |
| Deep Potholes | 19 (38%) |
| Shallow Potholes | 3 (6%) |
| Speed Bumps | 28 (56%) |
| Deep/Dangerous Events | 31 (62%) |
| Average Depth | 23.4 cm |
| Maximum Depth | 69.7 cm |
| Max Deviation | 92.3 cm |
| Avg Signal Strength | 3442 |
| Thread Frames (total) | 3030 |
| Thread Errors | 237 (7.8%) |

## Graph 1 — Timeline: Surface Deviation (cm)

Each point represents a detected anomaly. Blue line traces the deviation trend. Colour indicates detection type. Negative values = downward deviation (potholes/dips); positive = upward (bumps/raised sections).

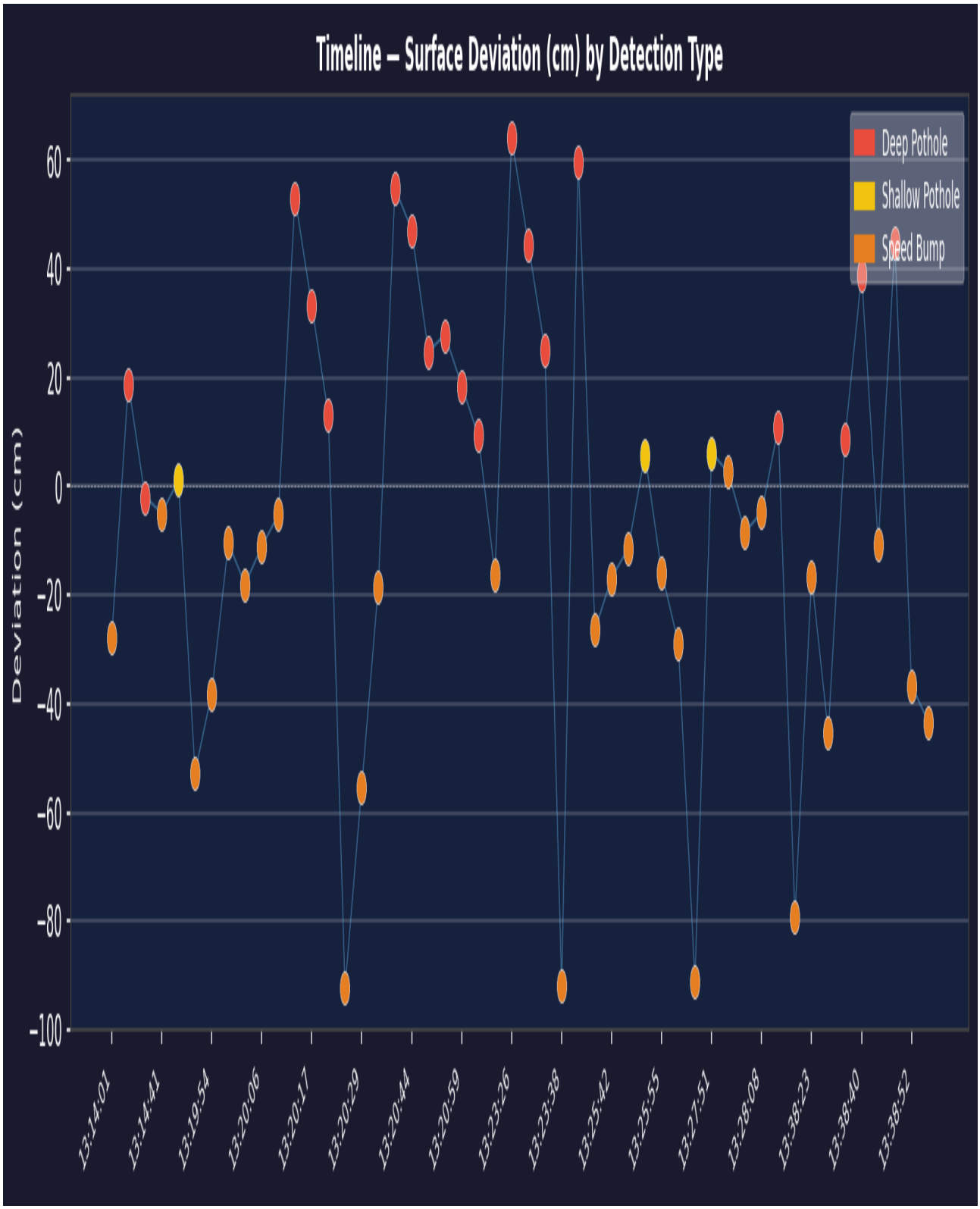


Figure 1: Deviation timeline across all 52 recorded events.

## Graph 2 — Timeline: Pothole/Bump Depth (cm)

Bar height = measured depth of each anomaly. Colour indicates severity: red = Deep/Dangerous, orange = Moderate, yellow = Shallow, grey = Noise.

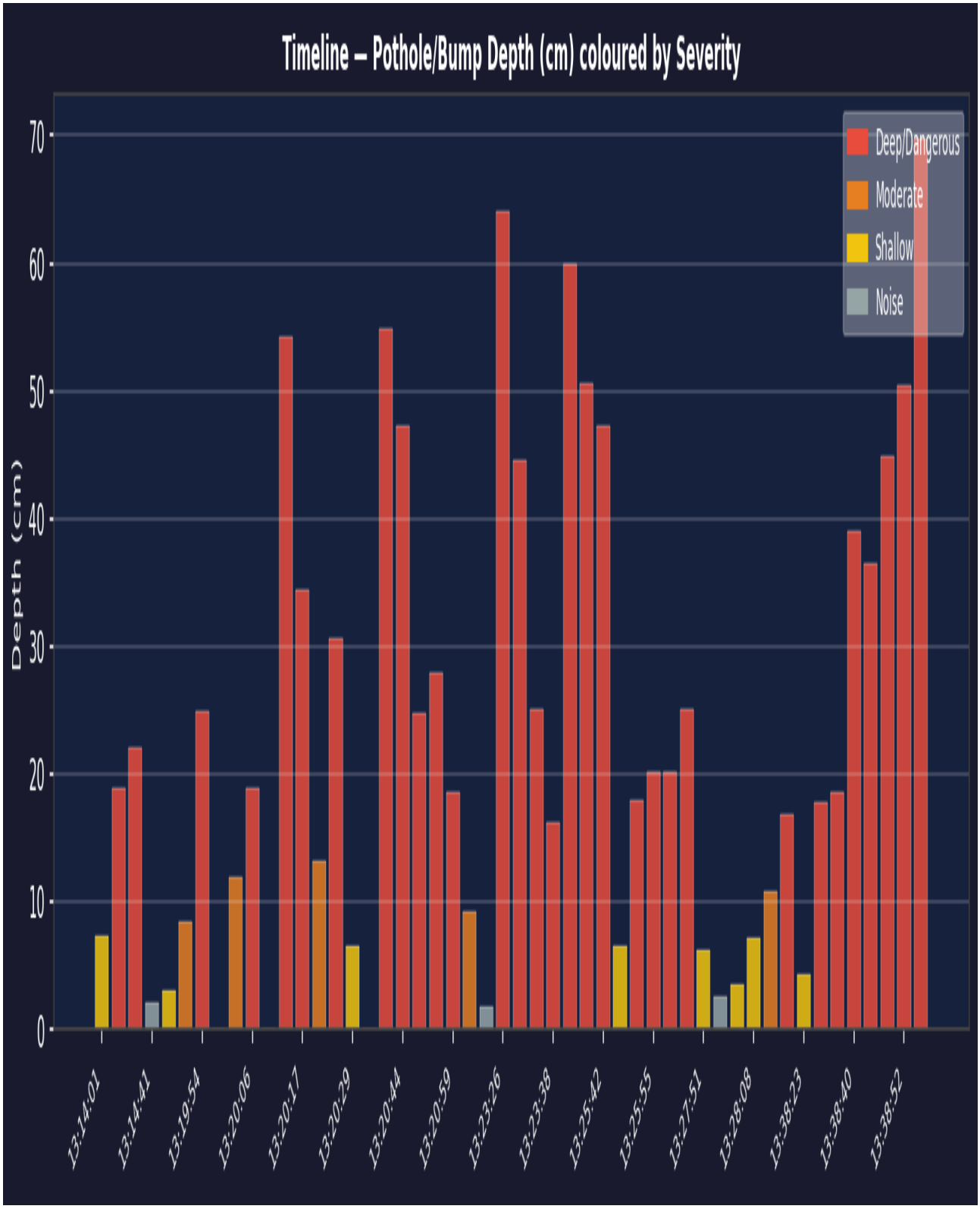


Figure 2: Depth profile with severity colouring.

## Graph 3 — Timeline: Signal Strength & Baseline Distance

Signal strength (teal) reflects sensor return intensity. Baseline (blue dashed) is the rolling reference distance in cm. Spikes in strength often correlate with solid or hard surfaces. A strong signal + large deviation = high-confidence pothole.

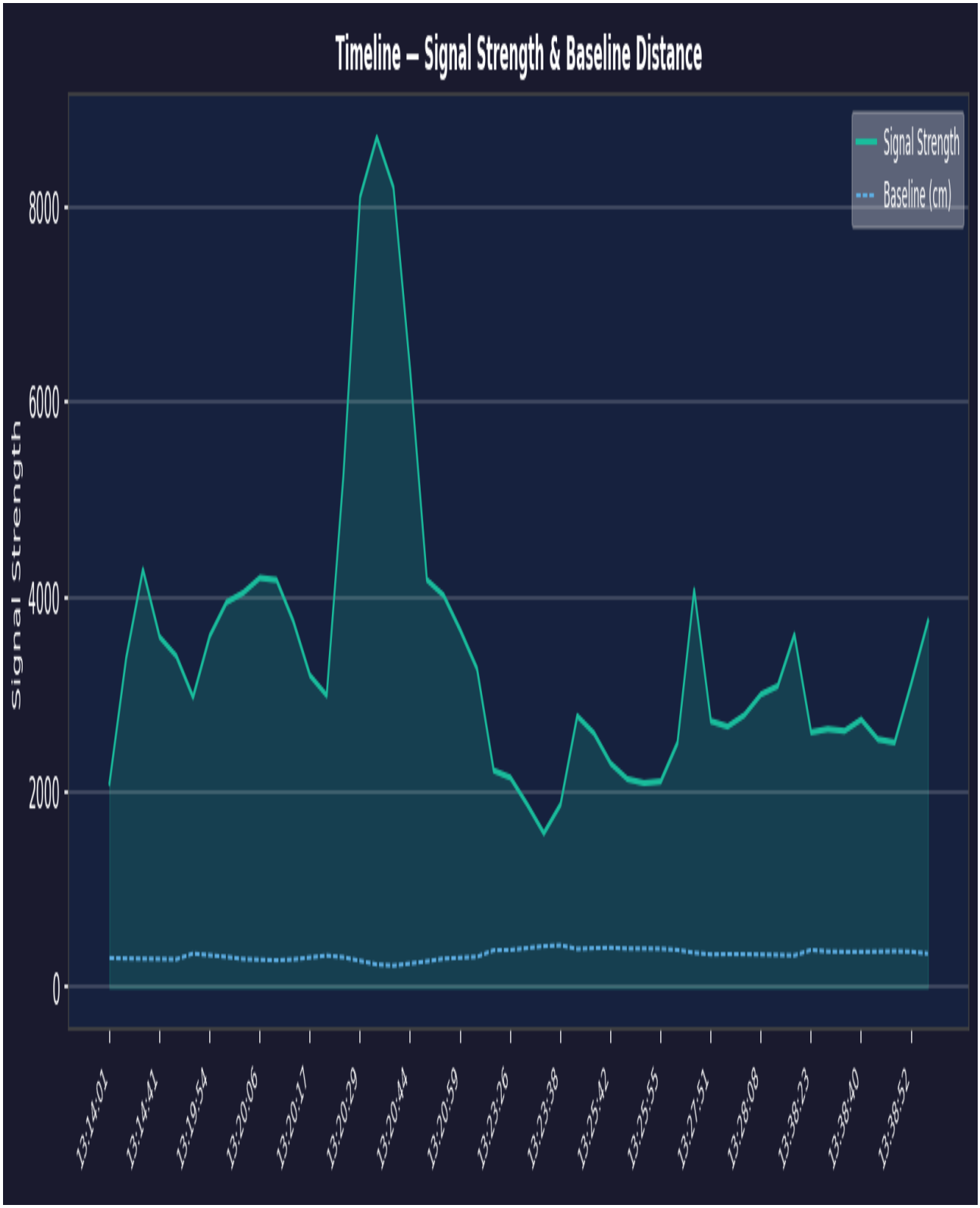


Figure 3: Signal strength and baseline distance over time.

## Graph 4 — Detection Type Distribution

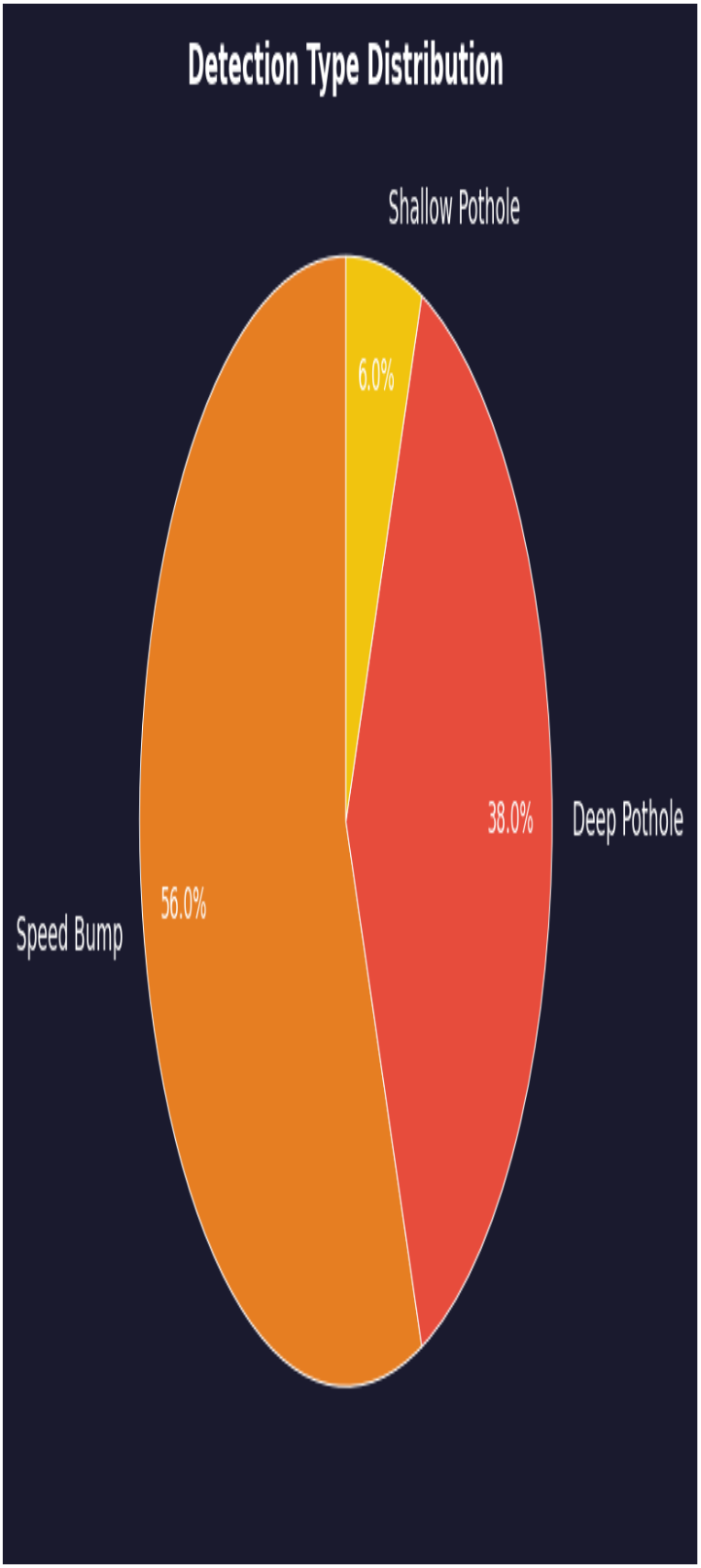


Figure 4: Proportion of detection types across the session.

## Graph 5 — Severity Level Distribution

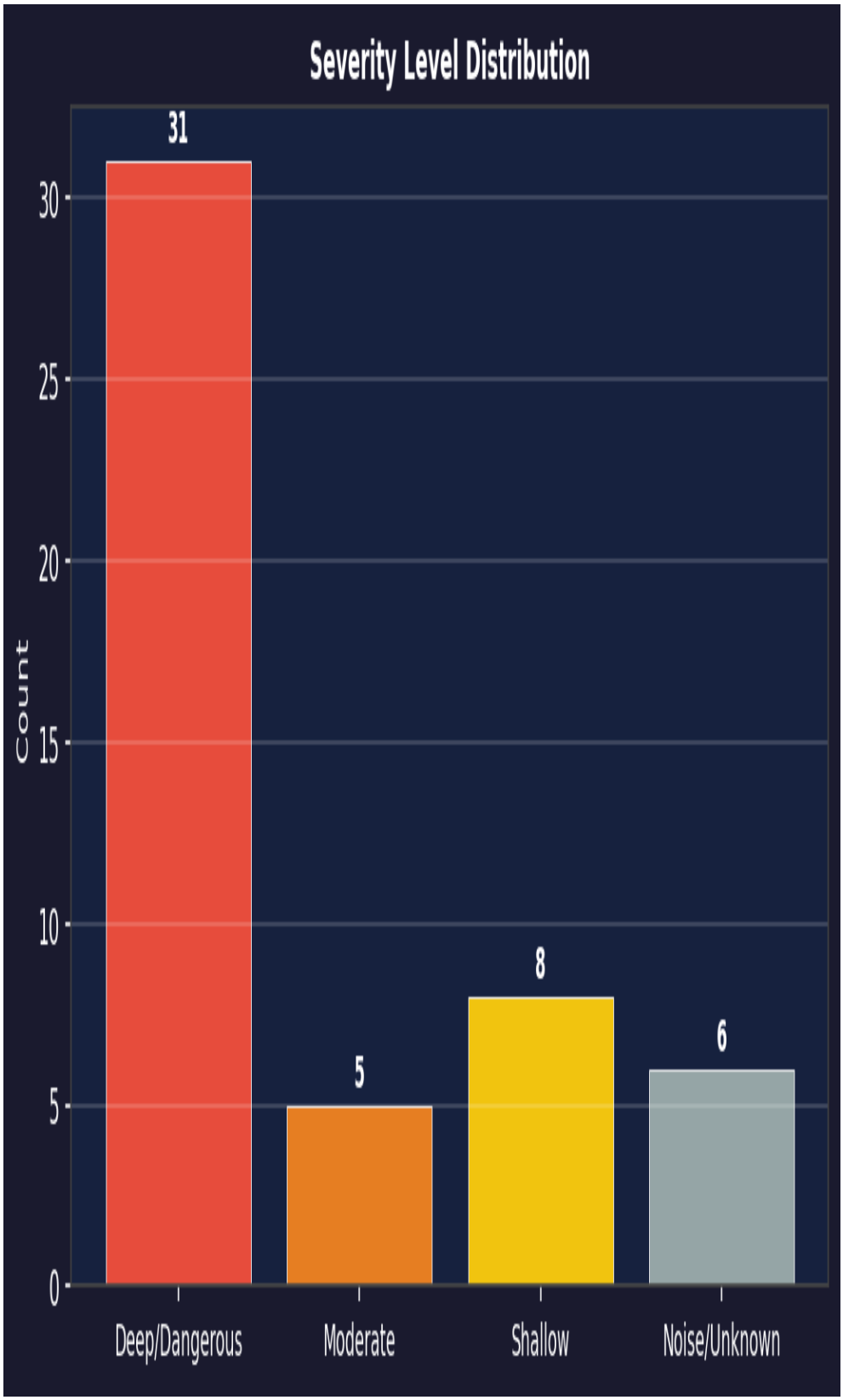


Figure 5: Count of events per severity category.

## Graph 6 — Anomaly Size: Length vs Width

Each point represents one detection, plotted by its estimated physical dimensions. Larger points in the upper-right represent wider, longer anomalies — potentially more dangerous.

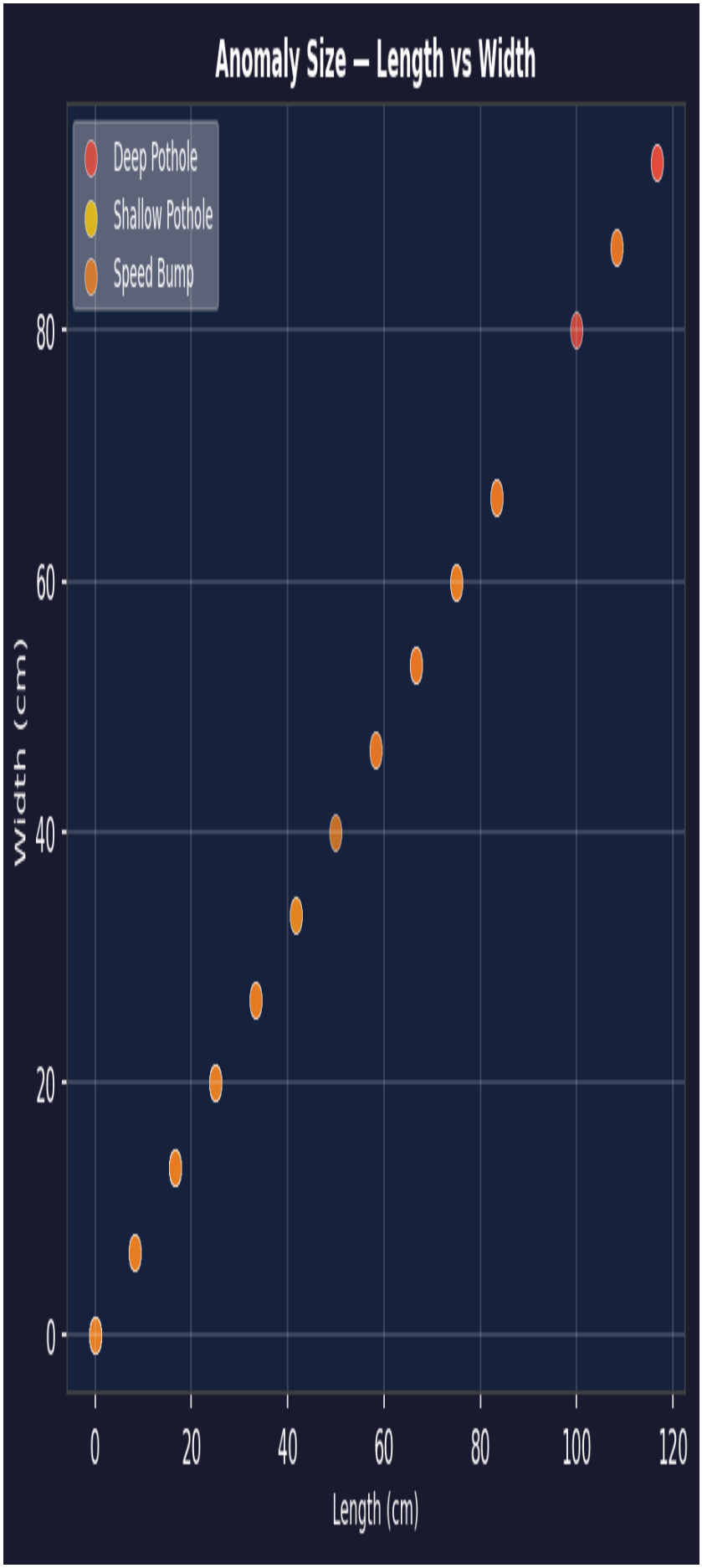


Figure 6: Length vs Width scatter, coloured by type.

## Graph 7 — Depth Distribution Histogram

Distribution of measured depths across all events. Yellow dashed = mean depth, teal dashed = median depth.

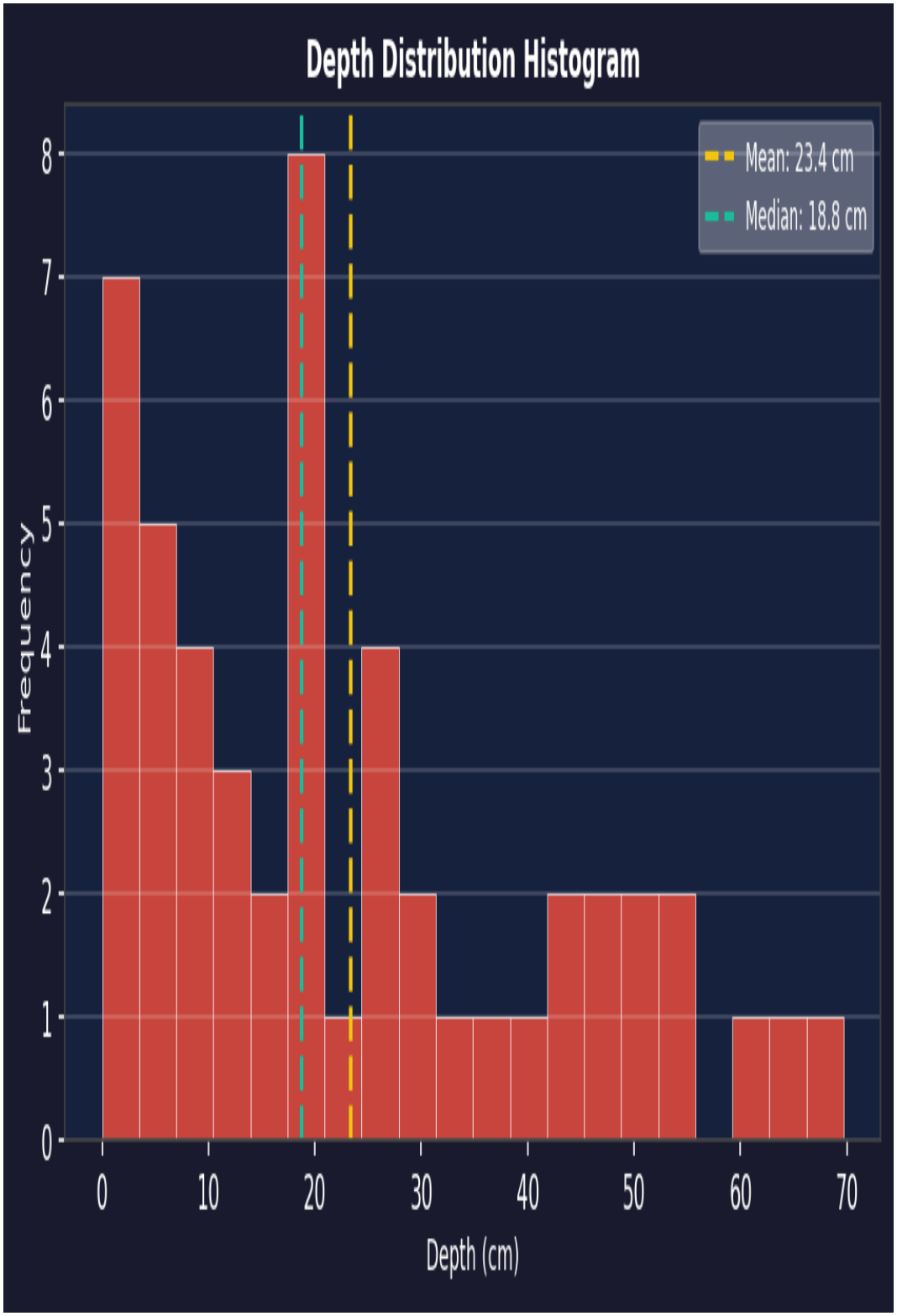


Figure 7: Depth histogram showing concentration of severe anomalies.

## Complete Detection Log — Chronological

All events sorted by time, including type, deviation, depth, size, severity, signal strength, and baseline distance.

| # | Time | Type | Dev (cm) | Depth (cm) | L×W (cm) | Severity | Strength | Baseline (cm) |
|---|---|---|---|---|---|---|---|---|
| 1 | 13:14:01 | Speed Bump | -27.7 | 7.3 | 25×20 | SHAL ■ | 2094 | 298 |
| 2 | 13:14:05 | Deep Pothole | +18.9 | 18.9 | 75×60 | DEEP ■ | 3379 | 294 |
| 3 | 13:14:15 | Deep Pothole | -1.9 | 22.1 | 25×20 | DEEP ■ | 4284 | 291 |
| 4 | 13:14:41 | Speed Bump | -4.9 | 2.1 | 0×0 | NOISE | 3593 | 289 |
| 5 | 13:15:11 | Shallow Pothole | +1.1 | 3.1 | 0×0 | SHAL ■ | 3396 | 285 |
| 6 | 13:19:50 | Speed Bump | -52.6 | 8.4 | 75×60 | MOD ■ | 2975 | 344 |
| 7 | 13:19:54 | Speed Bump | -38.1 | 24.9 | 33×27 | DEEP ■ | 3602 | 327 |
| 8 | 13:19:58 | Speed Bump | -10.4 | 0.0 | 0×0 | NOISE | 3947 | 310 |
| 9 | 13:20:02 | Speed Bump | -18.0 | 12.0 | 25×20 | MOD ■ | 4045 | 288 |
| 10 | 13:20:06 | Speed Bump | -11.1 | 18.9 | 8×7 | DEEP ■ | 4198 | 281 |
| 11 | 13:20:10 | Speed Bump | -5.2 | 0.0 | 0×0 | NOISE | 4177 | 275 |
| 12 | 13:20:13 | Deep Pothole | +53.2 | 54.2 | 33×27 | DEEP ■ | 3755 | 284 |
| 13 | 13:20:17 | Deep Pothole | +33.4 | 34.4 | 83×67 | DEEP ■ | 3198 | 304 |
| 14 | 13:20:21 | Deep Pothole | +13.2 | 13.2 | 100×80 | MOD ■ | 2993 | 324 |
| 15 | 13:20:25 | Speed Bump | -92.3 | 30.7 | 50×40 | DEEP ■ | 5243 | 306 |
| 16 | 13:20:29 | Speed Bump | -55.4 | 6.6 | 17×13 | SHAL ■ | 8106 | 269 |
| 17 | 13:20:33 | Speed Bump | -18.4 | 0.0 | 0×0 | NOISE | 8719 | 232 |
| 18 | 13:20:40 | Deep Pothole | +54.9 | 54.9 | 17×13 | DEEP ■ | 8202 | 220 |
| 19 | 13:20:44 | Deep Pothole | +47.2 | 47.2 | 67×53 | DEEP ■ | 6323 | 242 |
| 20 | 13:20:48 | Deep Pothole | +24.8 | 24.8 | 117×93 | DEEP ■ | 4178 | 264 |
| 21 | 13:20:55 | Deep Pothole | +27.9 | 27.9 | 17×13 | DEEP ■ | 4019 | 292 |
| 22 | 13:20:59 | Deep Pothole | +18.6 | 18.6 | 67×53 | DEEP ■ | 3661 | 301 |
| 23 | 13:21:03 | Deep Pothole | +9.3 | 9.3 | 117×93 | MOD ■ | 3266 | 311 |
| 24 | 13:23:23 | Speed Bump | -16.2 | 1.8 | 0×0 | NOISE | 2222 | 378 |
| 25 | 13:23:26 | Deep Pothole | +64.1 | 64.1 | 17×13 | DEEP ■ | 2149 | 381 |
| 26 | 13:23:30 | Deep Pothole | +44.6 | 44.6 | 67×53 | DEEP ■ | 1875 | 400 |
| 27 | 13:23:34 | Deep Pothole | +25.1 | 25.1 | 108×87 | DEEP ■ | 1579 | 420 |
| 28 | 13:23:38 | Speed Bump | -91.8 | 16.2 | 83×67 | DEEP ■ | 1874 | 429 |
| 29 | 13:25:31 | Deep Pothole | +59.9 | 59.9 | 8×7 | DEEP ■ | 2782 | 391 |
| 30 | 13:25:36 | Speed Bump | -26.4 | 50.6 | 42×33 | DEEP ■ | 2605 | 400 |
| 31 | 13:25:42 | Speed Bump | -16.8 | 47.2 | 8×7 | DEEP ■ | 2295 | 404 |
| 32 | 13:25:46 | Speed Bump | -11.4 | 6.6 | 25×20 | SHAL ■ | 2134 | 395 |
| 33 | 13:25:51 | Shallow Pothole | +5.9 | 17.9 | 42×33 | DEEP ■ | 2093 | 395 |
| 34 | 13:25:55 | Speed Bump | -15.8 | 20.2 | 67×53 | DEEP ■ | 2110 | 393 |
| 35 | 13:25:59 | Speed Bump | -28.8 | 20.2 | 33×27 | DEEP ■ | 2506 | 381 |
| 36 | 13:26:03 | Speed Bump | -90.9 | 25.1 | 8×7 | DEEP ■ | 4074 | 352 |

| # | Time | Type | Dev (cm) | Depth (cm) | L×W (cm) | Severity | Strength | Baseline (cm) |
|---|---|---|---|---|---|---|---|---|
| 37 | 13:27:51 | Shallow Pothole | +6.2 | 6.2 | 17×13 | SHAL ■ | 2729 | 335 |
| 38 | 13:28:00 | Speed Bump | +2.6 | 2.6 | 0×0 | NOISE | 2673 | 338 |
| 39 | 13:28:04 | Speed Bump | -8.4 | 3.6 | 0×0 | SHAL ■ | 2790 | 337 |
| 40 | 13:28:08 | Speed Bump | -4.8 | 7.2 | 17×13 | SHAL ■ | 3003 | 334 |
| 41 | 13:28:14 | Deep Pothole | +10.8 | 10.8 | 17×13 | MOD ■ | 3089 | 330 |
| 42 | 13:28:18 | Speed Bump | -79.2 | 16.8 | 108×87 | DEEP ■ | 3616 | 324 |
| 43 | 13:38:23 | Speed Bump | -16.6 | 4.4 | 0×0 | SHAL ■ | 2614 | 381 |
| 44 | 13:38:33 | Speed Bump | -45.2 | 17.8 | 75×60 | DEEP ■ | 2647 | 365 |
| 45 | 13:38:37 | Deep Pothole | +8.6 | 18.6 | 83×67 | DEEP ■ | 2628 | 362 |
| 46 | 13:38:40 | Deep Pothole | +39.0 | 39.0 | 25×20 | DEEP ■ | 2745 | 362 |
| 47 | 13:38:44 | Speed Bump | -10.5 | 36.5 | 58×47 | DEEP ■ | 2542 | 364 |
| 48 | 13:38:48 | Deep Pothole | +44.9 | 44.9 | 58×47 | DEEP ■ | 2510 | 368 |
| 49 | 13:38:52 | Speed Bump | -36.6 | 50.4 | 25×20 | DEEP ■ | 3117 | 363 |
| 50 | 13:38:56 | Speed Bump | -43.3 | 69.7 | 17×13 | DEEP ■ | 3756 | 343 |

# Analysis & Prevention Recommendations

## 1. Root Causes of Detected Anomalies

The sensor data reveals a high concentration of **Deep/Dangerous potholes** (depth > 20 cm) especially in the 13:20–13:26 time window, suggesting a stretch of severely deteriorated road. Speed bumps showed large negative deviations (up to –92 cm), pointing to unofficial or unmarked humps. Elevated baseline values (280–430 cm) in later segments indicate the vehicle was ascending or the road surface was rising relative to the sensor mount height.

## 2. Sensor / System Issues — Thread Errors (7.8%)

The dashboard showed **237 thread errors out of 3030 frames**. These likely stem from:
• USB latency or serial port timeouts causing sensor read failures
• Vibration-induced noise on the sensor mount at higher vehicle speeds
• Processing bottleneck in the local inference pipeline (Streak: 0/2, Cooldown: 3s)

**Prevention:** Use a hardware-buffered serial interface (e.g., FTDI chip). Mount the sensor on a vibration-dampened plate. Increase the cooldown window or use interrupt-driven reads instead of polling.

## 3. Road Infrastructure Recommendations

**Immediate (0–1 month):**
• Flag all *Deep/Dangerous* locations (depth > 20 cm, length > 50 cm) for emergency patching
• Install temporary warning signs or traffic cones near the 13:20–13:26 hotspot **Short-term (1–3 months):**
• Commission a professional road survey with ground-penetrating radar for sub-surface voids
• Apply cold-mix asphalt patch to moderate potholes (10–20 cm depth)
• Re-profile speed bumps to standard IRC 99:1988 dimensions to reduce impact **Long-term (3–12 months):**
• Full resurfacing of identified corridor with bituminous macadam (BM) overlay
• Install proper drainage channels to prevent future pavement failures due to waterlogging
• GPS-tag all anomaly locations and integrate into a Municipal Road Asset Management System **Sensor System Improvements:**
• Add GPS co-ordinates to each detection record for precise geolocation mapping
• Deploy redundant sensors (front + rear) to cross-validate detections
• Implement a Kalman filter on raw sensor readings to reduce false positives
• Log temperature (currently shown as 24.5°C) — high temp affects asphalt integrity

## 4. Severity Priority Matrix

| Severity | Depth Range | Recommended Action | Timeframe |
|---|---|---|---|
| Deep/Dangerous ■ | > 20 cm | Emergency repair / road closure | < 1 week |
| Moderate ■ | 10–20 cm | Schedule patching | < 1 month |
| Shallow ■ | < 10 cm | Monitor & seal cracks | < 3 months |
| Noise / Unknown | — | Review sensor calibration | Ongoing |